

LSTM Gate Equations

LSTM Equations

Symbol	Meaning
f_t, i_t, o_t	Forget, Input, Output gates
$\tilde{C}_t$	Candidate cell state
C_t, h_t	New cell state and hidden state
$W_\bullet, b_\bullet$	Weights and biases for each gate
$\odot$	Element-wise (Hadamard) product
$[h_{t-1}, x_t]$	Concatenation of previous hidden state and current input

$$\text{Forget Gate} \rightarrow f_t = \sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \quad (1)$$

$$\text{Input Gate} \rightarrow i_t = \sigma(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \quad (2)$$

$$\text{Candidate Cell State} \rightarrow \tilde{C}_t = \tanh(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t}) \quad (3)$$

$$\text{Output Gate} \rightarrow o_t = \sigma(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \quad (4)$$

$$\text{New Long-Term Memory} \rightarrow C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (5)$$

$$\text{New Short-Term Memory} \rightarrow h_t = o_t \odot \tanh(C_t) \quad (6)$$

$$h_t = \sigma(w_{ho_t} \cdot h_{t-1} + w_{xo_t} \cdot x_t + b_{o_t}) \odot \tanh(\sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \odot C_{t-1}) +$$

$$\sigma(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \odot \tanh(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t})$$

$$C_t = \sigma(w_{hf_t} \cdot h_{t-1} + w_{xf_t} \cdot x_t + b_{f_t}) \odot C_{t-1} + \sigma(w_{hi1_t} \cdot h_{t-1} + w_{xi1_t} \cdot x_t + b_{i1_t}) \odot \tanh(w_{hi2_t} \cdot h_{t-1} + w_{xi2_t} \cdot x_t + b_{i2_t}) \quad (7)$$

$$\begin{aligned}
h_{t-1} = & \sigma(w_{ho_{t-1}} \cdot h_{t-2} + w_{xo_{t-1}} \cdot x_{t-1} + b_{o_{t-1}}) \\
& \odot \tanh \left(\sigma(w_{hf_{t-1}} \cdot h_{t-2} + w_{xf_{t-1}} \cdot x_{t-1} + b_{f_{t-1}}) \odot C_{t-2} \right. \\
& + \sigma(w_{hi_{t-1}} \cdot h_{t-2} + w_{xi_{t-1}} \cdot x_{t-1} + b_{i_{t-1}}) \\
& \left. \odot \tanh(w_{hi_{2t-1}} \cdot h_{t-2} + w_{xi_{2t-1}} \cdot x_{t-1} + b_{i_{2t-1}}) \right)
\end{aligned} \tag{8}$$

$$\begin{aligned}
C_{t-1} = & \sigma(w_{hf_{t-1}} \cdot h_{t-2} + w_{xf_{t-1}} \cdot x_{t-1} + b_{f_{t-1}}) \odot C_{t-2} \\
& + \sigma(w_{hi_{t-1}} \cdot h_{t-2} + w_{xi_{t-1}} \cdot x_{t-1} + b_{i_{t-1}}) \\
& \odot \tanh(w_{hi_{2t-1}} \cdot h_{t-2} + w_{xi_{2t-1}} \cdot x_{t-1} + b_{i_{2t-1}})
\end{aligned} \tag{9}$$